



Review

Nano and micro architected cues as smart materials to mitigate recalcitrant pharmaceutical pollutants from wastewater



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H I G H L I G H T S

- Potential ecological and environmental effects of pharmaceuticals are reviewed.
- Removal of pharmaceuticals by an array of advanced methods are discussed.
- Different nano-architectures as adsorbents are outlined.
- Advantages and drawbacks of various removal processes are delineated.

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A B S T R A C T

Pharmaceuticals have been recognized for saving billions of lives, but they also appear as a novel group of environmental pollutants. The presence of pharmaceutically active residues in seawater, surface water, wastewater treatment plants, sludges, and soils has been widely reported. Their persistence in the environment for extended durations exerts various adverse consequences, such as gene toxicity, hormonal interference, antibiotic resistance, sex organs imposition, and many others. Various methodologies have been envisioned for their removal from the aqueous media. Different processes have been restricted due to high cost, inefficient removal, generation of toxic materials, and high capital requirement. The employment of nanostructured materials to mitigate pharmaceutical contaminants has been increasing during the last decades. The adsorptive nanomaterials have a high surface area, low cost, eco-friendliness, and high affinity for inorganic and organic molecules. In this review, we have documented the rising concerns of environmental pharmaceutical contamination and their remediation by applications of nanomaterials. Nanomaterials could be a robust candidate for the removal of an array of environmental contaminants in water.

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1. Introduction

The development of pharmaceuticals is a milestone in human achievements, and they played a significant role to improve life quality by curing millions of deadly diseases. During the past three decades, pharmaceuticals and their residues have evolved as a source of environmental contamination (Bottoni and Caroli, 2018; Rasheed et al., 2019a, 2019b, 2019c; Rasheed et al., 2020a, 2020b, 2020c). They have been found in surface water, groundwater, wastewater treatment plant, influents, effluents, and sludge, etc. Various antibiotics, antidepressants, anticonvulsants, lipid regulators, hormones, anti-inflammatories, and beta-blockers are among the highly discovered pharmaceutical-based pollutants (Rasheed et al., 2020). These contaminants are toxic, and they play a role to alter the properties of water and may be responsible to pose a high risk to human health and the ecosystem (Bilal et al., 2019; Rasheed and Nabeel, 2019). Various pharmaceuticals have variable degradation rates. In the environment, they may have complete to limited degradation. Most of them are not highly persistent, but even their nonstop small and significant amount added in the environment from variable sources made them pseudo persistent. In addition, the discharge of these pharmaceutically related toxic compounds into the main water matrices extremely threatened the equilibrium of natural ecosystem, thus posing serious health-related issues (Aguilar-Pérez et al., 2020; Ali et al., 2021; Bilal et al., 2017, 2021). Slight knowledge is yet reported about the long-term effect of these pharmaceutical contaminants on flora and fauna. These pharmaceutical wastes are responsible to cause chronic diseases, cancers, antibiotic resistance, endocrine disruptions, metabolic problems, imbalance of microflora (Cincinelli et al., 2015). These contaminants are affecting aquatic life by disrupting various physiological functions, reduction in fertility, damaging fish-spawning, accumulation of contaminants in tissues of the body, and eutrophication (Yaqoob et al., 2020). The quantification, identification, control, and elimination of these pollutants is the main concern to reduce their adverse effects. Different advanced techniques, such as GC/MS, LC/MS, and UPLC/MS made their determination easy and established that these contaminants are found in small amounts as $\mu\text{g/L}$ or ng/L (Lindsey et al., 2001) because of this they may escape from treatments and start accumulation in the ecosystem due to low volatile characteristics (Rasheed et al., 2019a). To date, 3000 biologically active molecules have been determined and quantified in the environment (Richardson and Ternes, 2014). The population is also contributing a lot to pollute the environment by these pharmaceutical molecules by releasing these compounds into our surroundings by flushing

the unused drugs, inappropriate drug disposal, and human and animal wastes (Cincinelli et al., 2015). So, there is the utmost need to remove these dangerous toxic pollutants from the ecosystem to prevent ecological damage and bad effects on human health (Nabeel and Rasheed, 2020; Rasheed et al., 2018; Rasheed et al., 2018). The wastewater treatment industry is focusing on developing sustainable solutions by research for efficient purification. Because conventional methods including chemical, biological, physical, and mechanical treatments are insufficient to eliminate or degrade these complex molecules/metabolites. Nanotechnology is an alternative to mitigate the pharmaceutical contaminants because these materials have nanoscale dimension ranges from 1 to 100 nm, along with large surface area, great reactivity, a high number of active sites, high porosity, great absorptivity, catalytic, photocatalytic, and redox properties (Thines et al., 2017; Yaqoob et al., 2020). Though nano-technology represents a modern tool to treat polluted water by designing nanomaterials that interacts with contaminants through binding, in-activation and degradation (Bilal et al., 2020a, 2020b; Ikehata et al., 2006; Saharan et al., 2014). In this review, we have reported the pharmaceutical existence and linked risks. Efficiency of wastewater treatment plants and other different removal technologies are reported. Efforts are carried out to report data for therapeutic, prevention, and degradation with a special focus on nanomaterials and detection of pharmaceutically active compounds.

2. Pharmaceuticals in the environment

Pharmacologically active compounds persist in aqueous media, are resistant to degradation, and responsible for causing harmful effects in living beings (José Rivera-Utrilla, Gómez-Pacheco, Sánchez-Polo, López-Peñalver and Ocampo-Pérez, 2013a, 2013b). Pharmaceuticals are not homogeneous molecules and they possess variable structural, biological, chemical, and physical characteristics. Pharmaceuticals have been developed by keeping in view the specific target in living beings. Differences exist between pharmaceutical contaminants and ordinary contaminants in many aspects having: (i) molecular Masses approximately < 500 Da however few molecules may have large (ii) huge structural variations and functionalities (iii) containing polar components and lipophilic characteristics and some may have mild solubility in water (iv) capability to stay in nature and remain active biologically (v) tendency to adsorb and distribution in living body systems (Kümmerer, 2009a; José Rivera-Utrilla, Gómez-Pacheco et al., 2013a, 2013b). Pharmaceutical consumption varies globally. Data regarding the consumption of pharmaceuticals in many countries is

available, but this available data is not sufficient (Van Boeckel et al., 2014). unregulated selling of drugs is happening in many countries so this also causes uncertainty for an estimate of the consumption and patterns of usage as well (Kümmerer, 2009a). In the united kingdom, almost 3000 pharmaceuticals registered but over 5000 counter-drugs are being sold (Defra, 2007). During 2001, almost 5000 drugs were registered in Germany, and from them, 2700 were >90% were consumed corresponding to almost 38,000 tons of active pharmaceutical compounds (Greiner and Rönnefahrt, 2003). The United States consumed 22,700 tons of antibiotics alone in 2001. Globally antibiotics consumption was 100,000–200,000 tons/annum in 2002 (Van Boeckel et al., 2014) and this quantity has multiplied with the time passage. One study estimated that by 2020 the usage of pharmaceuticals will extend to 4.5 trillion doses. By 2020 approximately 50% population of the world will be using >1 dose/person/day of medicine. Presently approximately 4000 pharmaceutically active ingredients are accessible in the market with 100,000 tons per year annual consumption globally (Weber et al., 2016). The socio-economic conditions influence the consumption of pharmaceuticals. In Greece during the years 2010–2014, the economic crisis changed the consumption pattern of pharmaceuticals particularly the usage of psychoactive drugs. Consumption pattern of pharmaceuticals also varies with seasons. The antibiotics consumption increases during winter season. Consumption pattern is also variable on basis of location, region and country. On large scale, the pharmaceuticals are used in agriculture zone and also to cure animals (Sarmah et al., 2006). Antibiotics, steroids, non-steroids, hormones and other pharmaceuticals are given to animals through topical and routes for prevention of infectious diseases and also for growth promotions (Sarmah et al., 2006). Rate of antibiotics consumption is far higher in animals as compared to human beings. Loading of pharmaceutical contaminants in environment from animal's cure is problematic because animal waste is not treated in comparison of human waste.

Active pharmaceutical compounds have been identified in aquatic systems during the 1980s. These active molecules have been grouped as antibiotics, antiepileptic, anti-inflammatories, antihistamines, beta-blockers, etc. (Bush, 1997). Antibiotics based contaminants are found in high amounts followed by analgesics but these patterns were found variable region-wise (Hughes et al., 2013). Mostly pharmaceutical contaminants have been detected in hospital wastes, effluents, sewage effluents, wastewater treatment plants (Heberer, 2002b; Rasheed et al., 2020). Traces of pharmaceutical contaminants have been found on every part of the planet even in the arctic environment due to human activities (Kallenborn et al., 2018). The spread of any pandemic may increase the use of specific drugs for a specific period that may lead to an increase in the load of that drug in wastewaters (Slater et al., 2011). Few such events have been reported but continuous observation is required of these events. The capability of pharmaceutical contaminants to adsorb on the solids, their biological transformation characteristics, and microbes present in the treatment process also contribute to environmental contaminations (Defra, 2007; Rasheed et al., 2019a, 2019b, 2019c). Before the excretion, the metabolic transformation of pharmaceuticals may or may not occur by gut micro-organisms or host enzymes (Mückter, 2006). The metabolic conversions may occur in two different phases. In phase I, different processes occur as hydrolysis, reduction, oxidation, leading to the modification of parent molecule into products that may be water-soluble with high polarity. While in phase II conjugation process may transform the parent molecule or the product obtained during Phase I. Hydrophilic derivatives are obtained during conjugation process which can be easily excreted. Either un-metabolized or metabolized both kinds of pharmaceuticals are emitted in the

environment directly through fecal matters. Pharmaceuticals which remains un-used or expired directly enters the environment through toilet, basins, drains and through household matters (Daughton and Ternes, 1999). According to a survey in England, it was observed that 66% population throw unused pharmaceuticals in household wastes, 12% population through them in flushes, only 22% give them back to seller (Bound and Voulvoulis, 2005). The pharmaceuticals present in waste materials of houses may be buried in landfills from where they may leach to groundwater and its contamination. Less polar and totally nonpolar pharmaceutical contaminants may adsorb onto wastewater treatment plant solids and sediments and may be removed with sludge. Application of these sludge materials to fields may release the pollutant into aquatic systems (Jones et al., 2005). Animal waste materials also contain hormones and different antibiotic molecules and when these wastes are applied to fields, they may migrate to groundwater. The largest source of pharmaceutical contaminants in the environment is urban-domestic discharge (aus der Beek et al., 2016; Bilal et al., 2020a, 2020b). The second-largest source of pharmaceutical contaminants is through effluents of hospitals while addition through animal excretion represents the third source. Pharmaceuticals are stable molecules but their Physico-chemical and biotic conversions are possible (Khetan and Collins, 2007). Environmental fate of pharmaceuticals can be predicted by understanding their metabolic process, biodegradability, conjugation, sorption, etc. Highly stable pharmaceutical molecules persist in the environment for a longer time. While pharmaceutical metabolites produced from reduction, hydrolysis, oxidation processes are susceptible to transformations that lead to their least stability (Löffler et al., 2005). Halogenation of various pharmaceuticals has been carried out to obtain high stability and bioavailability. So special attention is required for these halogenated molecules while formulation by keeping in view the groundwater legislation and environmental issues (Küster and Adler, 2014). In water bodies, they may undergo dilutions and chemical modification may change the parent drug molecules either completely or partially (Ihsanullah et al., 2016). Complete mineralization of parent drug molecules produces nitrates, carbon dioxide, sulfates, etc. (Kümmerer, 2009a). Pharmaceuticals may pass through three different environmental fates; the first is complete mineralization that may produce H₂O and CO₂ (Richardson and Bowron, 1985). Secondly, they may bound to environmental materials or retained in sludge while retaining the parent components. Usually, this is noticed for stable persistent molecules. Thirdly they may convert to further hydrophilic components that may remain persistent. The detailed transformation of the pharmaceutically active compounds has been illustrated in Fig. 1.

3. Toxic effects and risks of pharmaceuticals in the environment

Experiments show that environmental risk factor from pharmaceuticals depends upon the volume of consumption, their physical and chemical properties, and toxic effect on the ecosystem. Some factors affect the organism that necessarily takes into account risk analysis such as solubility of water, toxicity, persistence, bio-accumulation, and carcinogenesis. Pharmaceutical residue in the environment with low concentration cause chronic exposure on present microorganisms such as plants and animal. The particular result scales from metabolic alteration to hormonal imbalance. Even at very low concentration, some pharmaceutical has a very adverse effect on the environment and some pharmaceutical effect is similar like a human to other organisms (Kümmerer, 2009a, 2009b). Fig. 2 illustrated the adverse effects of antibiotics traces in

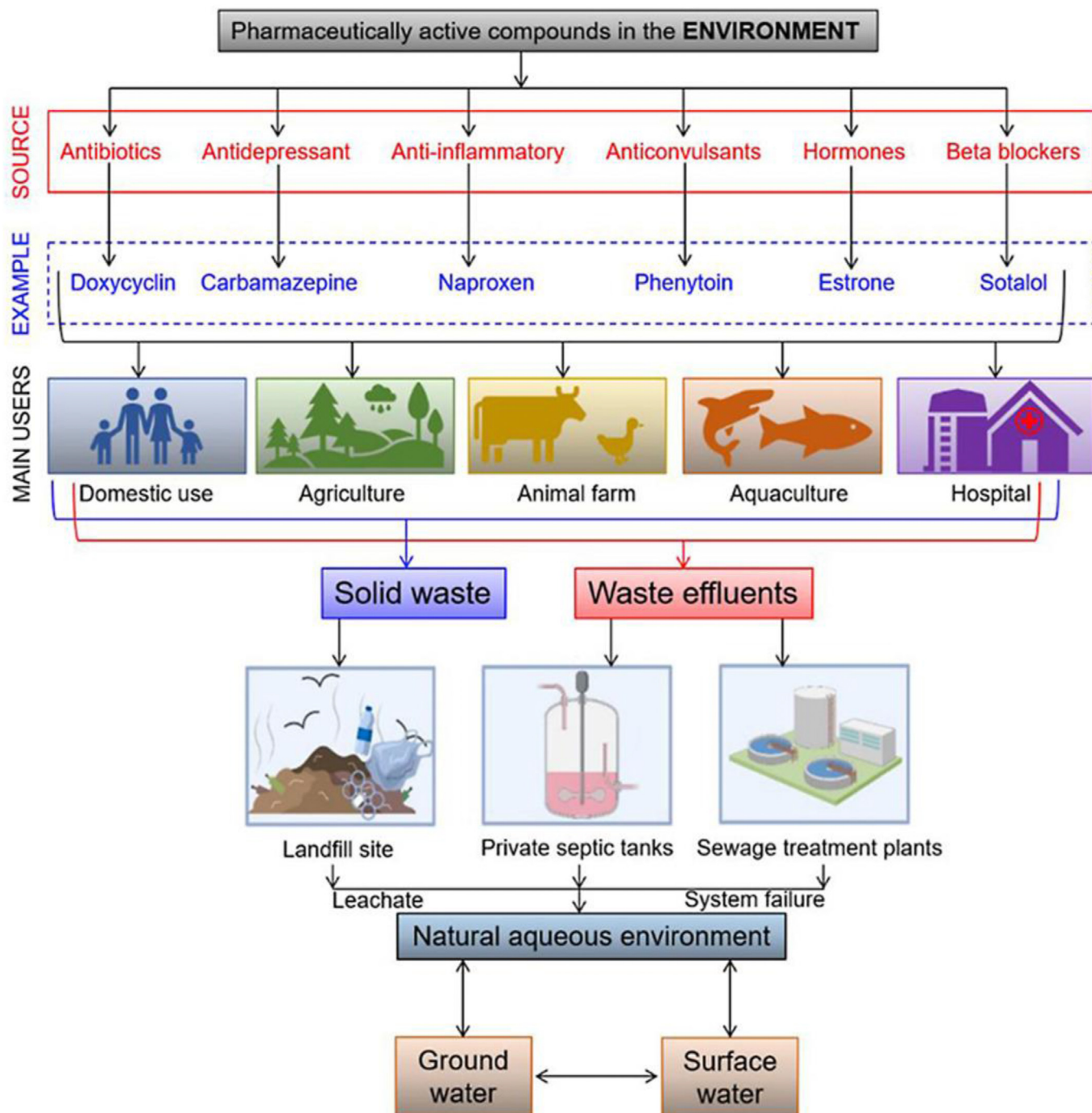


Fig. 1. Pharmaceutically active entities and their transformation routes to our water supplies and environment. Reproduced with permission from ref [Rasheed et al., 2020]. Copyright 2020, Elsevier Inc.

the environment to human health (Bilal et al., 2020a).

3.1. Toxicity and health risks in humans

Mostly the concentrations of pharmaceutical in ng/L are reported in drinking waters. The health risks of routine exposure to the low concentration of pharmaceuticals are not properly understood. Our knowledge is limited about the long term effect of such drinking water (Kimura et al., 2004; Polar, 2007). Moreover, a human can ingest these pharmaceuticals through their food like meat, vegetables, and fruits (Malchi et al., 2014; Prosser and Sibley, 2015).

Intake of these low concentration hardly any pharmaceutical exceeds limit during the whole life span. The value of therapeutic dosage every day would be greater than intake value during the whole life span (I_{70} stands on the expected life of 70 years). The interpretation of I_{70} is kept for particular compounds (Ternes, 2001). But, in environment presence of pharmaceutical are like a complex pool. So, serious questions arise about the long-term effect of a mixture of pharmaceuticals in drinking waters (Stackelberg et al., 2004). The evaluation of pharmaceutical risk is categorized by Critical Environmental Concentration (CEC) and Predictive No Effect Concentration (PNEC). We define PNEC as the concentration

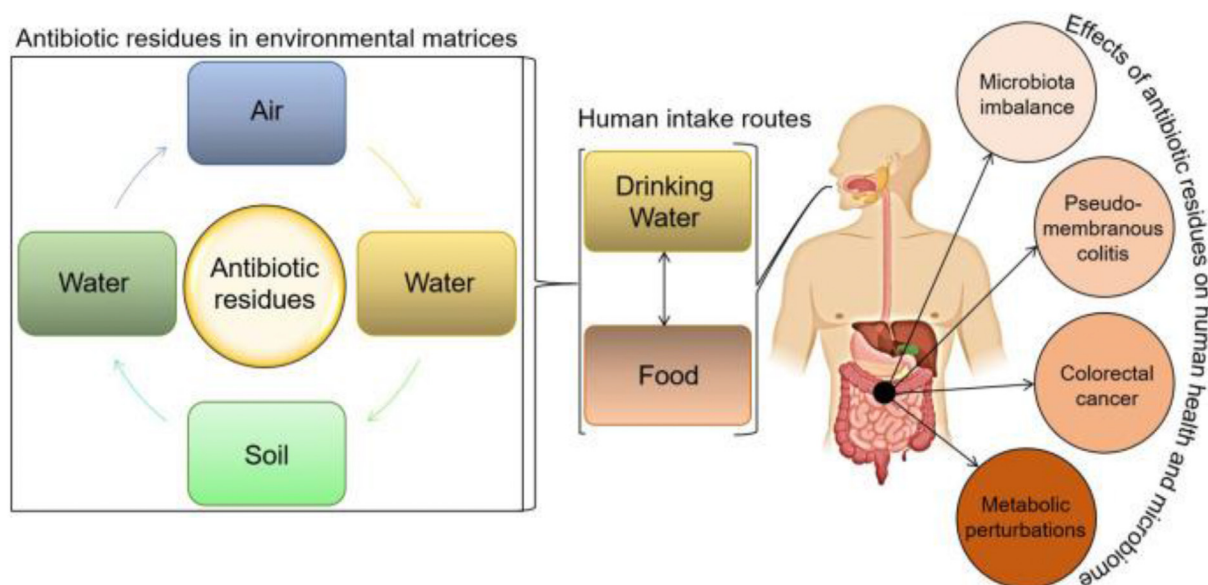


Fig. 2. Mechanistic framework of human-health related adverse effects of antibiotics traces available in different environmental matrices. Adopted from Bilal et al. (2020a) with permission from Elsevier. Copyright (2019) Elsevier B-V.

of pharmaceutical with no expected pharmacological effect on a particular organism. Ecological toxicity testing data determine the value of PNEC. Yves and Frederic (Frédéric and Yves, 2014) divide pharmaceuticals into three categories. Category 1 includes those compounds having no ecological toxicity hazards or having a hazard quotient less than one ($HQs < 1$). These compounds have a value of PNEC greater than $1 \mu\text{g/L}$ and are fewer ecotoxic effects. Category 2 includes those compounds having intermediate ecological toxicity hazard or has hazard quotient value in 1 and 1000. These compounds have a value of PNEC greater than 100 ng/L but lower than $1 \mu\text{g/L}$. Category 3 includes those compounds having HQs value greater than 1000. These compounds contain most ecological toxic compounds having a PNEC value lower than 1 ng/L . Toxicity and risk of same pharmaceutical varies place to place. Risk variations among south eastern-Spain, eastern-Sweden and northern Italy were determined in studies established on basis of the Swedish Environmental classification of levofloxacin and ciprofloxacin (Oldenkamp et al., 2014). Fig. 3 shows their impacts on human health and the aquatic environment for the above-mentioned locations (Oldenkamp et al., 2014). The risk of these pharmaceutical compounds also differs for another organism. So, a complete guideline is required for PNEC value and quality of water over a big number of terrestrial and aquatic species. Effect of diclofenac, fluoxetine, carbamazepine, and propranolol tried on many species (Kumar et al., 2016). Guideline value depends on survival rate which is about 95% for all tested species. Guideline value for diclofenac = $770 \mu\text{g/L}$, carbamazepine = $9.2 \mu\text{g/L}$, fluoxetine = $1.6 \mu\text{g/L}$ and propranolol = $14 \mu\text{g/L}$ were recorded (Fig. 4) (Kumar et al., 2016).

4. Pharmaceutical removal methods

4.1. Conventional treatment of drinking water

If drinking water contains pharmaceuticals, then it means that source water also contains pharmaceuticals. Multistage treatment of drinking water in conventional plants includes coagulation/flocculation, clarification, and filtration, and following this is the disinfection step. Further ozone oxidation, carbon (activated),

advanced oxidation processes (AOPs), and membrane filtrations are advanced methods that augment the treatment processes. All pharmaceuticals cannot be removed in the conventional plants when used for surface water (Adams et al., 2002; Ternes et al., 2002; Westerhoff et al., 2005). Therefore, sterilization or disinfection can remove only limited pharmaceuticals from drinking water. These two processes use ozone, chlorine, and chlorine dioxide. In the clarification process, suspended solid particles are removed by treating the source water to disrupt colloidal particles chemically, which assist their flocculation and enhance to settle them down. In a FeCl_3 clarification plant, pharmaceuticals that are hydrophilic in nature are included (dehydronifedipine, sulfamethoxazole, acetaminophen) (Stackelberg et al., 2007) and Coagulation of these compounds with (FeCl_3) gives rise to acid-base hydrolysis (Stackelberg et al., 2007). Clarification can only remove little quantity of caffeine, erythromycin, H_2O , and carbamazepine (Stackelberg et al., 2007; Ternes et al., 2002). The average removal of ethynylestradiol, caffeine, estradiol, estrone, androstenedione, and progesterone with Al and ferric salt-based coagulants was obtained (Stackelberg et al., 2007). The mitigation of sulfamethoxazole, meprobamate, pentoxifylline, carbamazepine, Dilantin, iopromide, diclofenac, gemfibrozil, triclosan, testosterone, acetaminophen, hydrocodone, trimethoprim, and naproxen, was approximately $<20\%$ with Al (Stackelberg et al., 2007; Westerhoff et al., 2005). Disinfection in the treatment plant acts as the most important and final step for drinking water and it has the ability to remove a remarkable amount of micro-organisms and pharmaceuticals (Hollender et al., 2009). The disinfection of drinking water is done by mixing chlorine which gives rise to the degradation of compounds, but some pharmaceuticals still persist after treatment (Gibs et al., 2007; Stackelberg et al., 2004). Chlorine dioxide is much better for oxidizing pharmaceuticals than chlorine due to its complete removal capacity of dimethylaminophenazone, phenazone, propiphenazone, and diclofenac (Huber et al., 2005). Glibenclamide, cyclophosphamide, and Gemfibrozil removal did not significant, and no mitigation was observed with biazifibrate, ibuprofen, carbamazepine, and diazepam. while with the help of ozonation, carbamazepine reduced significantly and concentrations of typical oxidant in ozonation are less as compared with chlorination. Some

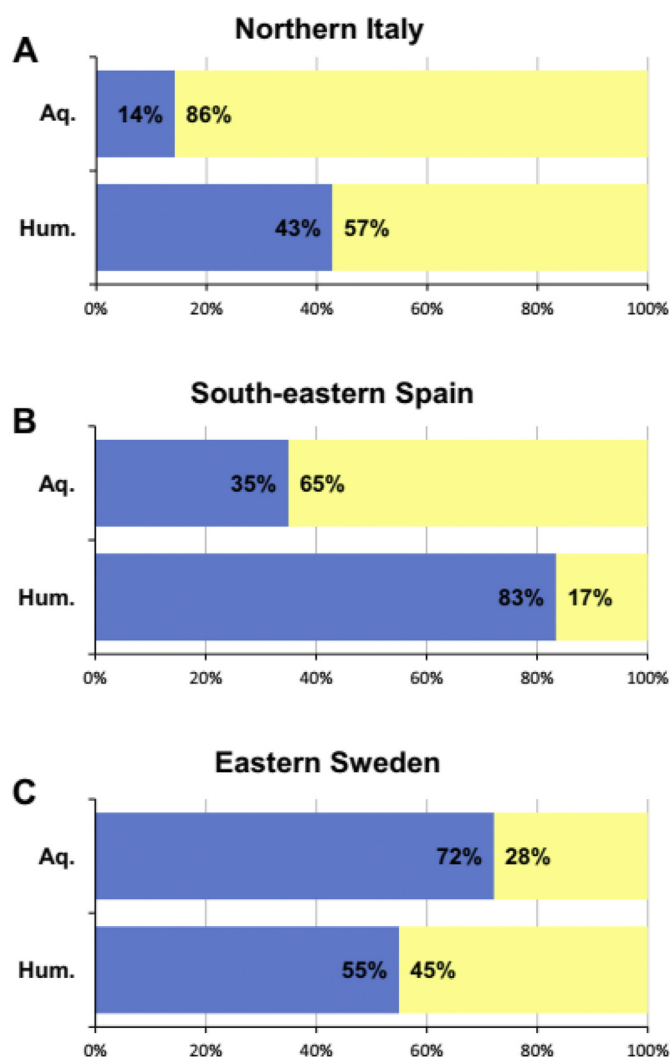


Fig. 3. Levofloxacin and ciprofloxacin relative impact on human health (HUM) and aquatic environment (AQ) in (A) northern-Italy, (B) southeastern Spain and (C) eastern Sweden. Blue and Yellow bars: relative influence of the prescription of one DDD of ciprofloxacin and levofloxacin, respectively. Reproduced with permission from (Oldenkamp et al., 2014). Copyright © 2014 Elsevier Ltd. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

important factors that affecting pharmaceutical oxidation are the pH value, ammonia concentration, dosage of free chlorine, existence, and concentration of suspended solids (Li et al., 2013). The pH of a solution takes an important part in the treatment of antibiotic chlorination. Different species of antibiotics are present in solutions, such as neutral, anionic, and cationic. Although many pharmaceuticals can be degraded remarkably with chlorination, for low concentration water systems its use is limited to prevent the concentration of organic carbon and ammonia (Yargeau and Leclair, 2008). It is also responsible for producing a huge amount of oxidation products and an insignificant mineralization amount. Toxic chloramines are responsible for producing Metopolo, acetaminophen, and sulfamethoxazole, while the various amounts of nonchloramine as by-product produced with diclofenac (Bedner & MacCrehan, 2006a, 2006b; Glassmeyer and Shoemaker, 2005). Acetaminophen leads to toxic 1, 4-benzoquinone imine, and N-acetyl-*p*-benzoquinone at the time of chlorination treatment.

4.2. Conventional wastewater treatment

Nowadays, the treatment of wastewater used both primary and secondary systems. The wastewater treatment plants (WWTPs) employed primary water screenings, biological filtration, aerobic and anaerobic digestion. These plants are not very suitable for the complete extraction of pharmaceuticals and other impurities. Most pharmaceuticals have not the ability to be efficiently absorbed in the activated sludge also some microbes not completely digest in the secondary treatment process (Göbel et al., 2005). Thus, these WWTPs are a suitable technique for the complete removal of these xenobiotics. listed the data of different studies of the removal range of pharmaceuticals by using WWTPs sewage treatment plants (STPs) and showing a large variation in percentage removal of these pharmaceuticals. These variations rely upon the design of the treatment plant, structure of pharmaceutical, and also environmental factors. Different factor controls the pharmaceutical decomposition such as plant construction, seasons, treatment technology, and conditions required for operating in WWTPs and STPs (Fent et al., 2006). Higher polarity usually increased the solubility of water and reduced the adsorption. Acidic pharmaceuticals like NSAIDs having low dissociation constants (pKa) values in the range of 4.1–4.9, biazafibrate (3.6 pKa), and At neutral PH the clofibrate acid ionized to their respective conjugate base (Ternes et al., 2004; Urase and Kikuta, 2005). However, basic pharmaceuticals and Zwitterionic also exhibiting significant adsorption and hydrophobic steroid hormones showing a solid-water distribution coefficients (log Kow) of 4 for adsorption (Golet et al., 2002; Ternes et al., 2002) Pharmaceutical environment depends on the temperature, pH value, compound physicochemical properties, light, and different diversity and activity of present micro-organism. The efficiency of removal is effectively dropdown during the winter season, while in summer the higher temperature has usually risen the microbial biodegradation.

4.3. Advanced water treatment

Organic compounds could be removed from water by using activated carbon adsorption. Some exceptions are very polar water-soluble iodinated contrast agents and antibiotic sulfamethoxazole (Westerhoff et al., 2005). Advanced oxidation processes are used to oxidize very small amounts of pharmaceuticals and other organic chemicals. But these AOPs show small efficiency removal as compared to ozone. Pore sizes of ultrafiltration and low-pressure microfiltration membranes are not appropriate for pharmaceuticals' permanent adsorption. MF and UF surfaces however can retain some hydrophobic compounds, results in short-term adsorption. MF and UF do not perform extra pharmaceuticals removal when applied in the membrane bioreactor system. However, when high-pressure membranes are applied in nanofiltration (NF) and reverse osmosis (RO) is very effective for pharmaceuticals removal from water (Rosman et al., 2018). Concentrated samples (brine) with a high amount of low-molecular-weight organic compounds and pharmaceuticals, like acetaminophen show problems in high-pressure membranes. Riverbank filtration (RBF) and soil-aquifer treatment (SAT) can be applied as additional steps for pre-treatment of drinking water or wastewater reclamation. These systems work as slow sand filtration along with high retention times. RBF and SAT are much effective at removal of very small amount of organic substances like pharmaceuticals by biotransformation and adsorption. Not much success has been made in controlling refractory compounds, like antiepileptic drugs (Drewes et al., 2003). Ultra-nano, microfiltration, and reverse osmosis is used for pharmaceutical removal. These systems require a

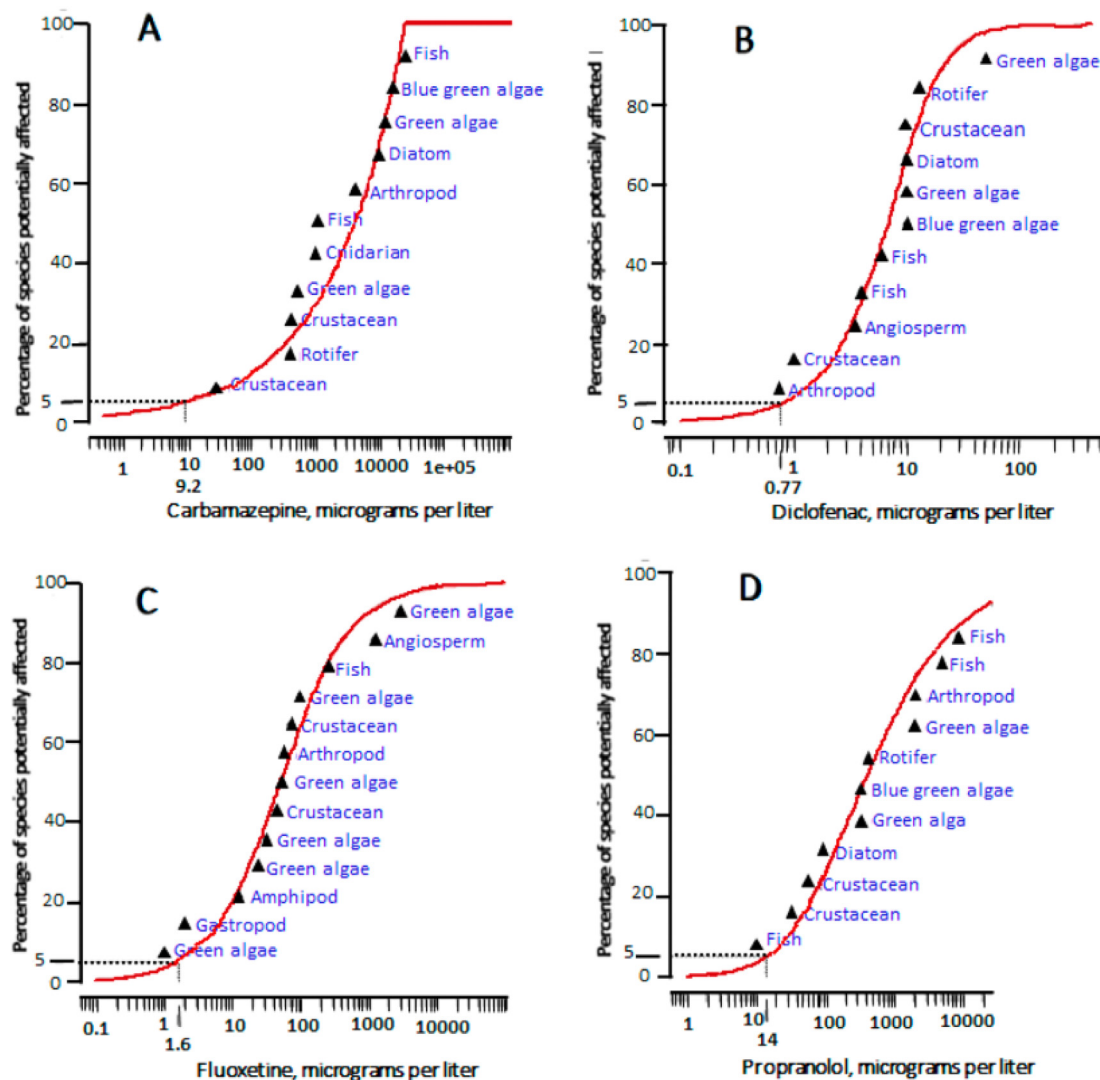


Fig. 4. Distribution and behaviour of different species toward various pharmaceuticals: (A) carbamazepine, (B) diclofenac, (C) fluoxetine, and (D) propranolol. Reproduced with permission from (Kumar et al., 2016). John Wiley and Sons © 2015 SETAC.

membrane that should be selectively permeable because systems are cross-flow filtrations (Radjenović et al., 2008). For reverse osmosis driving force is the difference between feed and filtrate of the membrane. In nano- and ultra-filtration primary force operational is ion repulsion because used membranes are charged. More than 85% of different NSAIDs are efficiently eliminated by nano-filtration and reverse osmosis (Radjenović et al., 2008). By using electrodialysis 99.7% ethinylestradiol was removed from urine. Diclofenac, ibuprofen, propranolol, carbamazepine, and ethinylestradiol were nicely removed from concentrated water with electrodialysis (Pronk and Kone, 2009). These physical separations combined with advanced electrochemical oxidation could improve pharmaceuticals removal (Sirés and Brillas, 2012). Reverse osmosis and nanofiltration removed higher levels of pharmaceuticals but ultrafiltration and microfiltration eliminated significant levels of pharmaceuticals (Snyder et al., 2007). In general, low pressure is used for nanofiltration (Zwiener, 2007).

4.3.1. Reverse osmosis-assisted removal of pharmaceuticals

It's a diffusion method that uses a selective semi-permeable membrane. Large molecules and ions are removed at low

pressure. This system is dependent on chemical nature, porosity, membrane type, membrane mechanical resistance, and membrane properties. High percent removal is the main benefit of this system. If suspended substances and large molecules clog membrane pores, it does not allow its use for wastewater having a large number of organic matrices. The effect of oxidizing agents and fouling is not suitable for large scale reverse osmosis (Watkinson et al., 2007). The major disposal concern is brine produced which contains toxic substances. To remove large amounts of dissolved materials from solutions reverse osmosis is used. Pressure applied is more than osmotic pressure (Boxall et al., 2011). Many pharmaceuticals (>90%) have been removed by reverse osmosis but diclofenac (55.2–60%) and ketoprofen (64.3%) show low removal (Verlicchi et al., 2010). Approximately >95% of propylphenazone, carbamazepine, diclofenac, naproxen, clofibric acid, and caffeine were extracted from drinking water in German water (Heberer, 2002a). The efficacy of removal was dependent on membrane pore size (Heberer, 2002a). Removal by reverse osmosis is directly related to physicochemical properties and pore size. Two tested membranes were cellulose acetate membrane (SC3100) and polyamide membrane (XLE) (SC3100) gave salt rejection value 94% with pore

molecular weight exclusion size between 200 and 300 amu. (XLE) gave 90% salt rejection value with pore molecular weight exclusion size <200 amu (Kimura et al., 2004). Overall, the XLE membrane showed higher pharmaceutical removal. Pharmaceutical rejection through XLE is correlated to the molecular weight of pharmaceuticals. However, in SC-3100 membrane polarity is the key factor for the rejection of pharmaceuticals. Both physiochemical properties and membrane material guide the mechanism of pharmaceutical rejection (Kimura et al., 2004).

4.3.2. Advanced oxidation methods

AOPs take place in an aqueous phase with very reactive species like HO_2 , $\text{HO}^\bullet$ and O_2 (Andreozzi et al., 1999). These radicals are generated by using ozone or hydrogen peroxide combined with UV radiations, or γ -rays or catalyst. Organic and inorganic pollutant oxidation is increased due to the high reactivity of these radicals. This is the reason for AOPs versatility (Esplugas et al., 2007). Oxidants used in AOPs are O_3 alone (Huber et al., 2003), O_3 combined with $\text{Fe}^{+2}/\text{H}_2\text{O}_2$ (Tekin et al., 2006), $\text{UV}/\text{H}_2\text{O}_2$ (Andreozzi et al., 2005), $\text{O}_3/\text{H}_2\text{O}_2$ (CFFH Zwiener and Frimmel, 2000), electron-Fenton degradation (Sires et al., 2007) and photo-Fenton oxidations (Pérez-Estrada et al., 2005). Many pharmaceuticals have been eliminated in a high amount (99%) by AOPs (José Rivera-Utrilla, Sánchez-Polo, Ferro-García, Prados-Joya and Ocampo-Pérez, 2013a, 2013b). AOPs have been used to naproxen (Méndez-Arriaga et al., 2008), paracetamol (Jallouli et al., 2017), moxifloxacin and clofibrac acid (Van Doorslaer, Dewulf, De Maerschalk, Van Langenhove and Demeestere, 2015), sulfamethoxazole (Pocostales et al., 2011), amoxicillin (Andreozzi et al., 2005), carbamazepine (Sun, Zeng and Lemley, 2013), tetracycline (Chen and Huang, 2011) ciprofloxacin (Jiang and Zhou, 2013), diclofenac (Pocostales et al., 2011) and ibuprofen (Jiang and Zhou, 2013). Researchers believe that AOPs decompose pharmaceuticals of wastewater efficiently. Photo-Fenton processes, photo-chemical AOPs, and ozone AOPs including $\text{O}_3/\text{H}_2\text{O}_2$, O_3/UV are more important than ozone alone. It is due to carbon-halogen bond breakage by photons and it increases the rate of hydroxyl free radical production (Ikehata et al., 2006). The main issue with AOPs is the toxic production of products; they can stay in remediated water. For example triclosan and triclocarban produce chlorodioxins and chloroanilines by advanced oxidation, both are toxic and lasting (Aranami and Readman, 2007). Naproxen produce products, more toxic than itself on oxidation (Isidori et al., 2005). Epoxycarbamazepine and other products are produced from carbamazepine, also includes carcinogenic and mutagenic acridine (Chiron et al., 2006).

4.3.2.1. Ozonation. Ozone (O_3) is a powerful oxidizing agent, it has the ability to act indirectly or directly. Ozonation is ozone use directly without irradiation and catalyst, while photoactivation and catalyst additional use are called AOP. Ozone reacts with electron-rich molecules and nucleophilic species and is electrophilic in nature itself (Sharma and Mishra, 2006). Direct oxidation takes place with molecules having carbon-carbon triple or double bond, aromatic rings, or at some sulfur, phosphorus, oxygen, or nitrogen functions (Ikehata et al., 2006). When hydroxide anion and ozone reacts in water hydroxyl radicals are formed (Andreozzi et al., 1999). Hydroxide initiates the sequence of reactions; hence pH shows an important role. pH increase raises reaction rate hence $[\text{HO}^\bullet]$ is also raised (Andreozzi et al., 2005).

Large numbers of studies are reported about ozone oxidation of pharmaceuticals (Klavarioti et al., 2009). After many reports analysis, it was finalized that ibuprofen and naproxen ozonolisation is best (Pal et al., 2010). Pharmaceuticals (macrolides, sulfonamides, and several acidic pharmaceuticals) in water are partially oxidized

by ozone (Huber et al., 2005). Ozonation decomposes sulfamethoxazole (amine), diclofenac (secondary amine group), and carbamazepine (carbon-carbon double bond) gives high removal rates (Huber et al., 2003). Diazepam, iopromide, and ibuprofen without expected functionalities show slow reaction rates on ozonation. To free water from pharmaceuticals, the majority used oxidation treatments is chlorination and ozonation. Ozone showed better results than chlorine dioxide in a variety of pharmaceutical degradation (Huber et al., 2005).

Based on reactivity in the chlorination and ozonation process pharmaceuticals are divided into 3 categories (Westerhoff et al., 2005). In the first categories, compounds included have low pK_a values, possess aromatic rings along with amine and hydroxyl function. They are removed (>95%) with chlorine and ozone. By deprotonation strong nucleophilic electron-rich species are formed and react fastly with electrophilic ozone (Pocostales et al., 2011). The second category shows not much oxidation (<20%) with chlorine and ozone both. These pharmaceuticals are aliphatics with the chlorine possessing functional moieties. The third category prefers ozone interaction but not chlorine. Examples are meprobamate, ibuprofen, and carbamazepine (Hahn and Schulz, 2007; Kumar et al., 2016; Westerhoff et al., 2005). All pharmaceuticals are not degraded by ozone as mentioned above.

Category 1 pharmaceuticals are successfully degraded by ozonation. However full and fast degradation and mineralization is not guaranteed for this category. For example ozonation of sulfonamide obtained more rapidly than ozonation of macrolides due to intensive reactivity of aromatic-rings than extensive saturated macrolide structure (Lin, Lin et al., 2009). As quinolone moieties are non-reactive towards ozone therefore fluoroquinolone are eliminated slowly (Vieno et al., 2007). Many factors affect the oxidation in both AOPs and ozonations, such as mode of operation, pharmaceutical concentration, oxidant dose, and water quality parameters like salinity effect, suspended solids, dissolved solids, and pH (Verlicchi et al., 2010). If the particulate matter amount is the same as the amount in secondary effluents then the removal of category 1 pharmaceuticals is not much influenced (Verlicchi et al., 2010). During oxidation soluble compounds decompose faster as compared to those adsorbed on the suspended matter; it is because of diffusion limits (Roberts and Thomas, 2006). Sulfamethoxazole, roxithromycin, erythromycin, diclofenac, indomethacin, clarithromycin, and Carbamazepine react fast with ozone, while ibuprofen, diazepam, and venlafaxine counter easy ozonation (Lester, Mamane, Zucker and Avisar, 2013).

Ozonation operations use an exact dose of ozone in the range of 5–15 mg/L chemical oxygen needs of wastewater determine this dose. The range of Contact-times is from approximately 15–30 min (Verlicchi et al., 2010). Cyclophosphamide ozonation in the wastewaters from hospitals produced approximately >97% expulsion and ozone needed was 256 mg/L (Ferre-Aracil et al., 2016). The pharmaceuticals which are present in pure water requires the 0.2 mg/L ozone for the oxidation, and bezafibrate needed only an amount 100 times less (Qiang et al., 2004). Antibiotics decomposition by ozonation, like lincomycin and spectinomycin, intensified with pH increase. This happened because of an increase in hydroxyl ($\text{OH}^\bullet$) radical amount produced from increase hydroxide amount at basic pH. The same conduct was also seen in the ozonation process of clofibrac acid (Andreozzi et al., 2003) but researchers agreed that in basic pH the rate increase was due to fast proton removal from clofibrac acid. While during the wastewater ozonation, particles decreased percent degradation of dopamine iopromide and iohexol (Zucker et al., 2015). Degradation drop had the order iopamide > iohexol > iopromide (Zucker et al., 2015). Greater than 90% of pharmaceuticals are removed by the average ozonation

process. But ng/L pharmaceuticals after ozonation stay in drinking water (Vanderford and Snyder, 2006). Ozonation is not effective completely (Vanderford and Snyder, 2006). As said earlier that carbamazepine and diclofenac oxidation products are dangerous. Moreover, ciprofloxacin decomposition is reduced due to quinolone moiety (Vieno et al., 2007). Finally, complex substances are degraded to simple substances that are biodegradable by ozonation in wastewater treatment, a step always not attained (Vieno et al., 2007).

Ozonation can be applied to effluent flow which fluctuates, but its implementation and maintenance are of great cost. The ozone generation needs regular energy input in it and has no advantage (Britto and Rangel, 2008). Operating cost increase by mass transfer of gas ozone (Britto and Rangel, 2008). At last ozonation process is affected greatly in presence of alkalinity, organic matrices, pH, temperature, chloride, and suspended solids (Andreozzi et al., 1999). For improving the efficacy and reactivity of ozonation H_2O_2 , irradiation and catalysts may be used at the same time. These compositions are AOPs. Some compounds are degraded by UV irradiations but not all (Rosenfeldt and Linden, 2004). UV irradiation converts hydrogen peroxide into hydroxyl radicals. These radicals are nonspecific and powerful and can be used for pharmaceuticals (Singh et al., 2014). Clofibric acid amount was decreased by 90% in 110 s, the initial amount used was 5–10 mg/L 17 W UV and 1020 mg/L H_2O_2 was applied for this purpose. These AOPs removed a large number of compounds, with or without ozone used. UV based degradation depends on contaminant molar adsorption capacity (ϵ), quantum yield (ϕ), and UV wavelength (λ) (López-Peñalver et al., 2010). At the given λ (S^{-1}) k_λ is degradation kinetic constant, ϵ_λ is compounds adsorption coefficient at the λ ($m^2 mol^{-1}$) used and E_λ represents light source emitted energy ($Einstein s^{-1} m^{-2}$). With this reagent combination different mechanism exists for AOPs. Some reactions are.

1. Hydroxyl release by photoexcitation and hydrogen peroxide cleavage by ozonation (eq 22.)
2. Ozone starts fast decomposition combined with UV and produces direct hydrogen peroxide synthesis in solution (Hernandez et al., 2002). Hydroxyls radicals are produced from this hydrogen peroxide.
3. Ozonation can be increased by employing the Hydrogen peroxide by perozonation, mechanism is the same as O_3/UV , enhanced by hydrogen peroxide. (Hernandez et al., 2002).

By most processes' diclofenac, naproxen, clofibric acid, and ibuprofen are not completely degraded. O_3/H_2O_2 combination works effectively to remove clofibric acid, diclofenac, and ibuprofen with mineralization 98% (Zwiener and Frimmel, 2000). For comparison of AOPs for cyclophosphamide and ciprofloxacin removal, the highest removal was produced by H_2O_2/O_3 and trimethoprim was best removed by O_3 .

4.3.3. Adsorptive removal

The attachment of Fluid molecules onto the solid surface is called adsorption. Four different steps are involved. 1. Transport of solute in the fluid phase. 2. Film diffusion. 3. Pore diffusion and 4. Adsorption on solid due to van Waals forces charge transfer complexes, hydrogen bonding, electrostatic attraction, in some cases strong chemical bonds and electrostatic forces produced by electron transfer (Homem and Santos, 2011). The efficiency of adsorption depends on adsorption surface, pore-volume, size, and porosity along with coverage of functional groups on the surface (Estevinho et al., 2007). For water treatment adsorption has advantages, like continuous and batch process sustainability can be

used on low adsorbate amount and low capital investment (Mohanty et al., 2006). Many inorganic and organic compounds can be removed by adsorption and produce less toxic substances than other removal methods. Adsorption is applicable to different types of wastewater and water samples with organic chemicals amount highly variable. Its limitation of other used methods. For pharmaceuticals removal of water solutions, adsorbents used are non-materials, polymer resins, mineral oxide, biochar and activated carbons. Rate constants and adsorption capacities of many adsorbents are integrated in this section. A dynamic biological adsorption equilibrium is observed, when desorption and sorption rates are equal, adsorbate total mass shift on adsorbent is zero, and adsorption capacity maximum is reached (Tahar et al., 2013). Isotherms models from isotherm experiments, explains the maximum adsorption capacity. To fit the adsorption isotherms, Freundlich and Langmuir's models are common (Langmuir, 1918). To fit data Dubnin-Radushkevich (Dubinin, 1947), Radke and Prausnitz (Radke and Prausnitz, 1972), Toth (1971), and Redlich-Peterson (Redlich and Peterson, 1959) models are also used. To show gas-solid phase adsorption on activated carbons, Langmuir adsorption was developed originally. Monolayer adsorption takes place in it. Localized sites are identical and equivalent in number. Even if the molecules are adjacent, adsorbed molecules show no lateral interaction or steric hindrance (Langmuir, 1918). This adsorption is homogenous and molecules show constant sorption energy of activation and enthalpy, and adsorbate shows no transmigration in the surface plane (Langmuir, 1918). Freundlich isotherm explains multilayer heterogeneous surface adsorption and distribution of adsorption affinities and heat are non-uniform (Freundlich, 1906a, 1906b; Freundlich, 1906a, 1906b). To show pharmaceuticals adsorption from water solutions Langmuir parameters and capacities are used. In reactions, kinetic studies are used to study adsorption rates. Many models are purposed to examine sorption pathways and to determine reaction rates, but for pharmaceuticals pseudo-first (Lagergren, 1898) and second-order (Ho and McKay, 1999) equations are the most used kinetics adsorption models (de Andrade, Oliveira, da Silva and Vieira, 2018). The adsorption capabilities of various materials have been demonstrated in Fig. 5.

4.3.3.1. Batch adsorption studies

4.3.3.1.1. Synthetically and commercially activated carbons.

In India the charcoal and carbon are used for the filtration process, it's old and deep from history. In 1500 BCE in Egypt, for medical adsorbent and purifying agent, carbonized wood was applied (Cheremisinoff and Morresi, 1978). The synthesis of activated carbon was initiated during 1900–1901 (Bansal et al., 1988). In 1930 in the United States reported the first implementation of activated carbon for the treatment of water (Mantell, 1968). A crude graphite form with a very porous structure, molecular level to visible cracks, and crevices pore size range is called Activated carbon (AC) (Hamerlinck et al., 1994). AC surface area, pore structure, and surface chemistry change with activation conditions and feedstock (Radovic, 2004). Adsorption capacities are related to pH, temperature, ionic strength, concentrations, and AC plus adsorbate properties. To clean large amounts of water volumes large AC amounts are needed hence not cost advantageous (Mohan and Pittman (2007)). An active area of research is to get low-cost adsorbents from agricultural and wood wastes (Essandoh, Kunwar, Pittman Jr, Mohan, & Mlsna, 2015). To adsorb aqueous phase pharmaceuticals AC is being mostly used (de Andrade et al., 2018). For pharmaceuticals removal, different applied activated carbon has been reviewed (Mansour et al., 2018). AC adsorption depends on pharmaceuticals' polarity. In pharmaceuticals, polar functional groups are common like sulfur and amide containing amine functions,

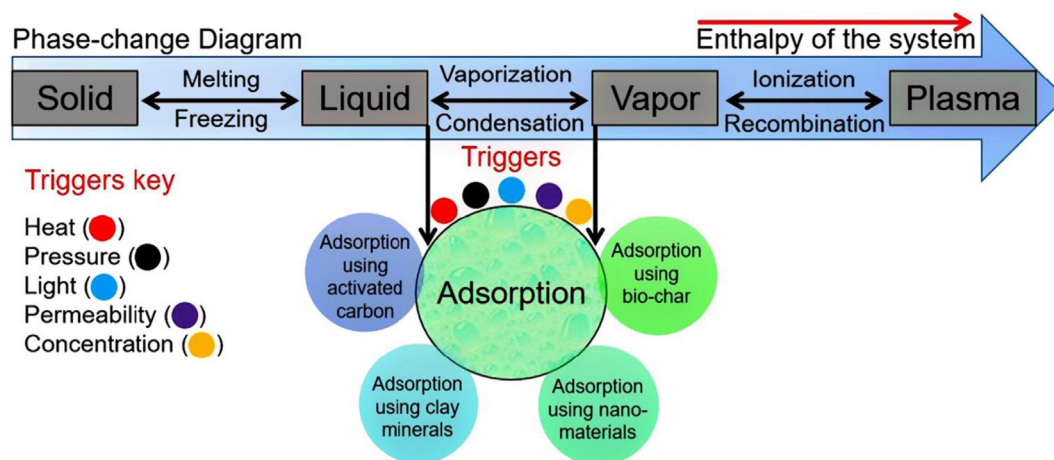


Fig. 5. Adsorption potentialities of different materials based on triggers and phase-changing treatment technologies (Reproduced with permission from ref [2019b]; an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)).

oxygen-containing groups, and N-heterocycles (Zwiener, 2007). One way to measure polarity is the octanol-water partition coefficient (K_{ow}) and is important to predict pharmaceutical efficiencies. A direct relation between the removal efficiency of adsorbent and $\log K_{ow}$ has been done (Westerhoff et al., 2005). Sulfamethoxazole ($\log K_{ow} \sim 0$) show low adsorption removal, while iopromide give low $\log K_{ow}$ ($\log K_{ow} = 0.48$) value (Ternes et al., 2002) and Carbamazepine ($\log K_{ow} > 3$) give high adsorptive removal (Hahn and Schulz, 2007). Compounds with polar heterocyclic groups like pentoxifylline pain killer and trimethoprim antibiotic show high removal value than given by K_{ow} value. It's due to interactions (ionic, hydrogen bonding, and π - π interactions) between protonated pharmaceuticals and activated carbon in solution (Zwiener, 2007). If pharmaceuticals are adsorbed on AC, they produce no toxic products but show high removal capacities and efficiencies (José Rivera-Utrilla, Sánchez-Polo et al., 2013a, 2013b). Equilibrium is faster obtained in hydrophobic compounds than hydrophilic compounds with AC (Snyder et al., 2007). For nitroimidazole (ronidazole, dimetridazole, tinidazole, and metronidazole) removal petroleum coke derived activated carbon and two commercial carbons (Sorbo and Merck) were studied (Rivera-Utrilla et al., 2009).

Carbon graphene layers π delocalized electron density and nitroimidazole aromatic ring dispersion is responsible for Nitroimidazole adsorption. Forces responsible for adsorption of nitroimidazole aromatic ring and carbon graphene layers are π - π interactions. Electron donating groups in nitroimidazole and carbon surface favor removal. In adsorption, electrostatic Adsorbent-adsorbate forces play no major role. Hydrophobic forces guide adsorption and carbons chemical properties dictate adsorption extent. Adsorption capacities and Carbon's oxygen content are directly related and increase with a decrease in nitroimidazole size. Commercial AC adsorption of metronidazole is pH related very much (Baig et al., 2019; Carrales-Alvarado et al., 2014). At pH 12, 390 mg/g the highest Langmuir capacity was achieved (Carrales-Alvarado et al., 2014). Metronidazole molecules at pH 12 were neutral and carbon's material was negatively charged. Adsorption of metronidazole increase when pH increases from 10 to 12. It shows that main adsorption interactions are dispersive and hydrophobic. This trend is supported by the fact that at pH 12, the solubility of metronidazole decreases due to high electrolyte amount and results in increased hydrophobic forces between metronidazole molecules and carbon surface. AC treated with nitric acid contains more phenolic groups and carboxylic acid groups

than commercial AC. Due to it adsorption was decreased at pH 7, while ionic strength and temperature exerted slight effects on adsorption. Adsorption Increase in wastewater indicates promotion by the electrolyte. For adsorption of sulfamethoxazole, Fe_2O_3/CeO_2 was loaded to AC from a commercial palm kernel shell (Akhtar et al., 2011). Powdered AC was able to absorb a high amount of sulfamethoxazole as compared to Fe_2O_3/CeO_2 blended carbon and temperature changes the adsorption and initial amount Tetracycline shows very high absorption on AC from petroleum coke, made active by KOH. This AC was very porous (2800–2900 m^2/g). At 30°, 40°, and 50 C° maximum Langmuir adsorption capabilities were 898, 962, and 1122 mg/g respectively and pH ~3 showed maximum adsorption. Tetracycline sorption is affected by pH because at different pH various ionic forms of tetracycline present (Zhu et al., 2014). Hydrothermal carbonization solid product is hydrochar. It was made active in a step with Ferric chloride to porous magnetic matrices for the sorption of tetracycline (Zhu et al., 2014). Magnetized carbon shows a high surface area than hydrochar and it showed 25 mg/g Langmuir adsorption capacity at 30 C° (Zhu et al., 2014). The rate of adsorption of the tetracycline on the sludge derived adsorbents follows intraparticle diffusion control, hence surface diffusion part is not important (Ocampo-Pérez et al., 2012). When granular activated carbon was applied for amoxicillin adsorption high Langmuir adsorption capacity of 222 mg/g at pH = 4.98 was obtained (Putra et al., 2009). This capacity is affected by pH change. Four times more amoxicillin is adsorbed on granular activated carbon than bentonite clay, (at pH 2.31 approximately 54 mg/g on the bentonite) per unit of adsorbent weight (Putra et al., 2009). Optimum adsorption pH and activated carbons pH (4.91) coincides and in this range neutral form of amoxicillin exists. At $\sim pH_{pzc}$ similar results of amoxicillin adsorption were (Dutta et al., 1999). Amoxicillin forms zwitter ions and hence both positive and negative groups exist at this pH. Hence electrostatic forces increase between ammonium and carboxylic groups of amoxicillin and adsorbate. Amoxicillin amine groups and AC carbonyl group can also interact as previously explained (Putra et al., 2009).

Two ACs were obtained by the use of K_2CO_3 and KOH, which activated *Albizia lebbeck* seedpods in a microwave oven (Ahmed and Theydan, 2013a). Both adsorbed cephalixin effectively with 137 and 119 mg/g maximum Langmuir adsorption for potassium hydroxide (KOH) and potassium carbonate (K_2CO_3) altered carbons. Endothermic adsorption was observed in Potassium hydroxide-modified carbon while exothermic adsorption was exhibited by potassium carbonate modified carbons. When *Siris* seed ACs

modified with KOH and K_2CO_3 were used for metronidazole adsorption, comparable findings have been obtained (Ahmed and Theydan, 2013a; 2013b). AC (NORIT ROX 0.8) is commercially available, it was modified by nitric acid oxidation and for ciprofloxacin, sorption heat-treated activated carbon was used (Carabineiro et al., 2011). Heat-treated carbon at 900 °C was applied for ciprofloxacin adsorption and presented of 311 mg/g Langmuir capacity. It was more than HNO_3 modified commercial AC (190 mg/g). Ciprofloxacin AC adsorption was found very much pH-dependent. When $pH = pK_{a1}$ of ciprofloxacin maximum adsorption is noted. When ciprofloxacin was adsorbed on date palm leaflet activated carbon, the role of pK_{a1} and pK_{a2} and pH was noted. When $pH = pK_{a1}$ (El-Shafey et al., 2012), adsorption was maximum. Sulfamethazine sorption increased 6 times (from 3.07 to 17.24) when $FeCl_3$ modified AC was used (Liu et al., 2017). Carbon's pore volume, zeta potential, and surface area were increased by $FeCl_3$ action, hence adsorption capacity was maximum (Liu et al., 2017).

Norfloxacin and acetaminophen sorption was analyzed on ACs from cattail fiber (Liu et al., 2013) and animal hair. Second-order kinetics is shown by these ACs, but adsorption of acetaminophen was faster than norfloxacin with an initial amount of 0.1 mmol/L 95% removal within 30 min at 25 °C was obtained. Filtrasorb F400 is a commercially available AC. Salicylic acid show exothermic adsorption on it. By Langmuir equation adsorption it presents monolayer and homogeneous adsorption (Otero et al., 2004). AC (351.0 mg/g) adsorbed more salicylic acid than on two resins (non-ionic) for Sephabeads SP206 45.2 mg/g and Sephabeads SP207 81.6 mg/g respectively. Ibuprofen adsorption capacity and kinetic on AC cloth was investigated. AC cloth surface was improved using NaOCl, or it was treated thermally (700 °C under N_2). It was concluded that the adsorption and kinetics were pH-dependent (Guedidi et al., 2017). As pH decreases adsorption increases and maximum adsorption done at pH 3.0. Basic carbonyl sites were formed on carbon cloth by heat at 700 °C in an inert nitrogen environment which increased adsorption. This treatment removes lactone functions and carboxylic acids; hence enough increase in carbonyl groups. Substrate becomes slightly basic by this treatment. Upon re-exposure of the heated substrate to dioxygen in air produce oxygenated moieties chiefly carbonyl moieties on aging. Some studies show adsorption of ibuprofen at pH range 3–4 (Bui and Choi, 2009). At high-temperature *Cyclamen persicum* tuber was activated under nitrogen conditions with residence time low (Jodeh et al., 2016). 22 mg/g was the maximum adsorption capacity of diclofenac this condition. Adsorption was affected by temperature, pH, and concentration. As the diclofenac amount increases adsorption increases but decreases with pH and temperature rise. For diclofenac sodium salt adsorption optimum pH is 2, less than its pK_a (4.20) because under this environment conjugate acid form is present which decreases water solubility. For adsorption of diclofenac, naproxen, ketoprofen and ibuprofen olive waste-cake based AC was used (Baccar et al., 2012). Adsorption was affected by pK_a , pK_{ow} , and pH, but the temperature had no effect. The presence of chloro group (electron-withdrawing) enhances adsorbate ability for π -electron-rich sites of adsorbents and results in an increase in adsorption. Chloramphenicol (CAP) and tetracycline (TC) were removed nicely by bamboo charcoal in the fixed-bed column and batch modes (Liao et al., 2013). The mass transfer mechanism was dominated by surface diffusion. CAP and TC adsorption capacity increased by increasing bed height and decreased by flow rate and influent amount increase (Liao et al., 2013). Often powdered and granular ACs for pharmaceuticals adsorption are applied. But for specific conditions and adsorbates new AC development is of great cost. So recent work focus is to develop adsorbents low cost, easily regenerated, and obtained from readily available and local resources (Ahmed, Zhou, Ngo and Guo, 2015).

4.3.3.1.2. Biochars. By International Biochar Initiative (IBI), Biomass carbonization produces solid matrices called biochar. They are being employed as both adsorbents and fuel (Mohan et al., 2014a, 2014b). They were being applied to remove pharmaceuticals, pesticides, polynuclear aromatics, phenols, dyes, and inorganic from water (Asmaly et al., 2015a, 2015b; Blaisi et al., 2018; Ihsanullah et al., 2015; Ihsanullah et al., 2020; Mohan et al., 2014a, 2014b; Zubair et al. (2021)). Biochar adsorption is affected by factors like temperature, pH, preparation condition, humic acid, ions, groups on the surface of the biochar's, and in its solid framework and pharmaceuticals structure. The low cost of Biochars made them versatile and they can replace granular ACs and water treatment cost can be decreased (de Andrade et al., 2018). Plant wood (oak stem, oak wood, douglas fir) (Dewage et al., 2018; Mohan et al., 2014a, 2014b), agricultural wastes (wheat husk, corn cob, rice husk) (Mohan et al., 2018), grasses (Essandoh, Wolgemuth, Pittman Jiang and Zhou, 2017), sludge, algae, tea residue (Vithanage et al., 2016), and municipal waste (Ahmad et al., 2014), were applied for biochar production. Many biochars and modified biochars were used for pharmaceuticals sorption from aqueous solutions. For example tea waste biochars (Rajapaksha et al., 2014), pinewood biochars (Dubey et al., 2010), wheat straw (Jia et al., 2018), rice husk (Liu et al., 2012), maize straw (Jia et al., 2018). Critical work was done on tetracyclines and sulfonamides removal from aqueous solution using biochars (Peiris et al., 2017). The major theme of this work was to develop a better contaminant extraction method by understanding the characteristics and sorption mechanisms. Pinewood biochar was produced by fast pyrolysis at 425 °C. The reactor was continuous and auger-feed, it was used for salicylic acid and ibuprofen removal (Essandoh et al., 2015). Salicylic acid showed maximum adsorption at pH 2 and ibuprofen at pH 3. Fast pyrolysis high surface area chars gave reasonably high capacities. When adsorption per unit area surface of different adsorbents was related for removal of ibuprofen and salicylic acid these chars were best. This factor can go over 100. These biochars are less carbonized. Water (H_2O) penetrates inside the internal hydrophilic reagents and moves into solid places inside. So, to have high adsorption, adsorbents do not stick to surfaces but penetrate. So, permeation take place and adsorbent is taken to solid structures under pore surfaces. So, adsorption does not take place at pore surfaces. These biochars have higher BET surface area for pharmaceuticals removal and thus are effective and efficient like other adsorbents (Essandoh et al., 2017a, 2017b). Biochars produced by fast pyrolysis for short residence time at low temperatures (425–500 °C), show reasonable oxygen content. They have less surface area. Despite it, they have many oxygenated functional moieties above and below surfaces of pores (Mohan et al., 2014a, 2014b). Dissolved adsorbate and water penetrate in solid biochars network, not limited to pore structures and adsorbates remain within the solid structure. Adsorbents with a low surface area can be excellent due to their property to swell in water and allowing more space for adsorbents. Variation in hydrogen bonding between adsorbate and adsorbent functional groups is a key factor for salicylic acid and ibuprofen adsorption. Adsorption of sulfamethazine was studied on tea waste biochars (Rajapaksha et al., 2014). When temperature increased from 300 to 700 °C, adsorption also increased. In case of tea waste activated biochars cation exchange, cation- π and electron donor-acceptor played key role. Sulfamethazine adsorption and effect of biochar production temperature was also investigated (Zheng et al., 2013). Production temperature increase caused increase in adsorption due to increase in surface area. Biochars obtained from various biomass sources can show different adsorptions. For example maize straw biochars (MBC600) gave the maximum capacity of 4.32 mg/g and wheat-straw biochar (WBC600) 3.06 mg/g as given in (Jia et al., 2018). Pharmaceuticals sorption is also affected by ions presence.

On maize straw derived biochar oxy-tetracycline adsorption is affected by metal ions (Jia et al., 2013). Sorption was increased by Cu^{+2} ions (0.2 nM), are 16.4%, 66%, and 7.5% at pH 3.5, 5.5 and 7.5 respectively. But from acidic solution, Pb^{+2} (0.2 nM) slightly decrease the oxy-tetracycline adsorption. Cd^{+2} , Zn^{+2} , K^{+} and Ca^{+2} had no effect on this adsorption process (Jia et al., 2013). Biochars modification can enhance the sorption capacity of pharmaceuticals on biochars. These modifications are steam activation, impregnation, heat treatment, and chemical modification (Ahmed, Zhou, Ngo, Guo and Chen, 2016). Chemical and specific surface properties are enhanced by these modifications and contaminant sorption capacity is increased (Ahmed et al., 2016; Khalid et al., 2018). Modified biochars' high removal performance and low cost are a good replacement for carbon nanotubes, AC, and graphene (Ahmed et al., 2016). Biochars modifications are delayed below. At 300 and 700 °C, steam activated biochars and nonactivated biochars of *Sicyos angulatus* L. were obtained and applied for adsorption of sulfamethazine (Rajapaksha et al., 2015). Biochars produced by steam activation at 700 °C had greater surface area compared to nonactivated biochar. Langmuir adsorption capacity of 38 mg/g was shown by SMT's on biochar prepared at 700 °C. The main interaction between biochar and SMT were π - π electron donor-acceptor (EDA) between biochar's π -electron rich graphene surface and protonated aniline ring of SMT. SMT has many moieties for hydrogen bonding including protonated nitrogen. Solution pH is responsible for acceptor donor interactions. Sorption in an acidic medium depends on the protonation of SMT and at high pH adsorption is less because SMT forms anions and electrostatic interaction between negative regions on biochar and SMT takes place (Rajapaksha et al., 2015). Humic acid loaded cotton, corn, and rice straw-based biochars, adsorbed enhanced sulfamethoxazole and low sulphanilamide in comparison of parent biochars (Lian et al., 2015). Fast pyrolysis rice husk biochars alkali and acid modified, were applied to remove tetracycline from the aqueous system (Liu et al., 2012). Biochars physiochemical character was changed due to these modifications. Langmuir adsorption potentials were raised from ~17 mg/g for un-modified to ~59 mg/g and ~23 mg/g for alkali and acid modified biochars, respectively. π - π interactions were enhanced by modifications. Adsorption capacity enhancement is concerned partly with increase surface and porosity of structure. Adsorption is not significantly affected by initial solution pH (Liu et al., 2012).

4.3.3.1.3. Clays and soils. Clays are phyllosilicates, naturally existing structures, and have a composition of tetrahedral and octahedral sheets in 2:1 or 1:1 proportions. Few clay examples are montmorillonite (2:1), smectites (2:1) and kaolinites (1:1). Normally, these are applied for pharmaceuticals sorptions. Clays are used due to their high ion exchange capacity and chemical stability. Chemical nature and pore structure is responsible for clays adsorption capacity (Babel and Kurniawan, 2003). Ionic pharmaceuticals with positive charges are well adsorbed, if clay structure has a negative charge. Removal of pharmaceuticals can be compared and in some cases higher than activated carbons (de Andrade et al., 2018). Adsorption of amoxicillin on bentonite is pH-dependent to a large extent (Putra et al., 2009). Surface complexation and cation-exchange are the possible interaction for sorptions and take place between protonated silanol/aluminol of bentonite and carboxylate anions. Bentonite cations of low valence interlayer interact with protonated functional groups of amoxicillin. On bentonite surface, alumina and silica surface sites are protonated at a solution ($\text{pH} < \text{pH}_{\text{pzc}}$) of bentonite (Putra et al., 2009). Promethazine hydrochloride, a cationic pharmaceutical is absorbed by K10 montmorillonite and at pH 7.5 gave a maximum potential of 326 mg/g. Adsorption potential decreases with a rise in temperature and this phenomenon confirms the exothermic type

of adsorption. This adsorption occurs by physisorption (Gereli et al., 2006). Promethazine hydrochloride adsorption on smectite which is iron-rich, with temperature rise first decreases than increases. The maximum adsorption capacities at $17 > 37 > 25$ °C. Giles classification confirms that C type isotherm is followed by adsorption curve, suggesting, that adsorbate has constant affinity for adsorbent (Seki and Yurdakoç, 2009). Ciprofloxacin is quickly adsorbed as compared to tetracycline by kaolinite (Li et al., 2011). These antibiotics ammonium forms, adsorb through the exchange of cations with Na^{+} and K^{+} present in the interstices of clay materials (Zhang et al., 2011). Montmorillonite, SAZ-1 ($S_{\text{BET}} = 65 \text{ m}^2/\text{g}$) ($\text{Ca}_{0.88}\text{Na}_{0.01}\text{K}_{0.05}[\text{Al}_{2.96}\text{Mg}_{1.32}\text{Fe}^{+3}_{0.18}\text{Ti}_{0.02}][\text{Si}_{7.86}\text{Al}_{0.14}]\text{O}_{20}(\text{OH})_4$) was employed for the adsorption of ciprofloxacin (Wang et al., 2010). Stoichiometric exchange took place between interlayer clay cations and ciprofloxacin and gave in neutral and acidic solutions an adsorption ability of 330 mg/g (Wang et al., 2010). Illite, rectorite, and montmorillonite were also used for ciprofloxacin adsorption and gave capacities of 33, 135, and 395 mg/g respectively (Wang et al., 2011). Important factors in sorption were electrostatic interactions and hydrogen bonding. Mineral's cation exchange-capacity (CEC) and adsorption capacities were directly related. For montmorillonite, rectorite and illite CEC order was $1.23 > 0.41 > 0.11 - 0.12$ mequiv/g respectively. In XRD expansion of basal layer spacing and FTIR characteristics, bands shift of protonated amine (1390 cm^{-1} [ketone (1630 cm^{-1})] shows that ions of ciprofloxacin intercalated with tilted alignment in illite, rectorite and montmorillonite during adsorption. The prominent mechanism in sorption was cation (Wang et al., 2011). For removal of ciprofloxacin through adsorption process the Sodium montmorillonite clay (MMT) was employed and confirmed by FTIR and XRD analysis (Wu et al., 2010). In solution at pH (3–8) higher CIP sorption was obtained, less than CIP's $\text{pK}_{\text{a}2}$ (8.74). CIP's cationic and zwitterions forms were adsorbed on MMT. After CIP uptake, interlayer d -spacing expanded which shows an interlayer adsorption, proved, intercalation of CIP took place (Wu et al., 2010).

Trimethoprim was adsorbed on K10 montmorillonite, and for isotherm, it produced L type Giles classification curve. It indicated that adsorbate and solvent have no competition for surface sites (Bekçi et al., 2007). An important role is played by the pH of the solution. Up to pH 6.3 adsorption increases and then an abrupt decrease in potential takes place. Protonated trimethoprim formation may be the reason at less pH, that can be adsorbed simply by cation exchange into the clay (Bekçi et al., 2007). Adsorption of trimethoprim on montmorillonite KSF showed huge removal near pH 5. Montmorillonite KSF is responsible for pH lowering of the solution because it is acid-activated and has pH of ~3.3 in suspensions. Montmorillonite KSF's cation exchange capacity also governs the role of pH (Bekçi et al., 2006). A major role is played by CEC in the adsorption of pharmaceuticals on clays. CEC of montmorillonite is higher than kaolinite, therefore montmorillonite adsorbs more Nalidixic acid (24 mg/g) than kaolinite (1.03 mg/g) (Wu et al., 2013). The pH also affects the pharmaceutical's sorption on soil and the presence of organic carbon. The purpose of both chemical and physical adsorption of clays is to increase removal efficiencies (R. Zhu et al., 2016). Main modification treatments are acid treatment, thermal treatment, and organic modification of clays. Thermal treatment can make clay more hydrophilic by dehydroxylation and dehydration. Texture and dispersive properties are also affected by thermal treatment. Calcination of bentonite (Verde-Iodo) at 500 °C increases the sorption capacity of caffeine ($8.78 \mu\text{mol/g}$) as compared to pristine bentonite ($3.27 \mu\text{mol/g}$) (Maia et al., 2017). Clay's acid treatment can alter the surface charge and also influence surface area and pore structure (Zhu et al., 2016). Organic compounds are well adsorbed by organic clays. These are obtained easily from parent mineral clays by using organic cations. Other

used organic-based modifiers are nonionic surfactants, cationic polymers, ionic liquids, chelating agents, and organic modifiers of more than one type (Rasheed et al., 2020; Zhu et al., 2016). These reagents replace inorganic cations by intercalating between clays layers and hence interlayer distance is enlarged. Kaolinite modified by cetyltrimethylammonium bromide (CTAB) results in higher adsorption potential of diclofenac (45 mmol/kg) as compared to pristine kaolinite that gives no adsorption almost (Sun et al., 2017). In gravel, organic matter, sand, and mineral clay composition of soil differ widely. Hence soil adsorption by soil is pH-dependent, widely variable, and complex (Verlicchi and Zambello, 2015). Adsorption of sulfamethazine on *Nicollet*, *Harps*, *Clarion-1*, *Clarion-2* and *Clarion-3* soil decreases with an increase in pH. Sorption coefficient values (K_d , (L/Kg) decrease for *Nicollet* (from 3.77 to 1.07), *Harps* (from 3.91 to 1.16), *Clarion-1* (from 0.58 to 0.23), *Clarion-2* (1.22–0.49) and *Clarion-3* (from 2.52 to 1.05) by rise in pH rise from 5.5 to 9.0 (Lertpaitoonpan et al., 2009). *Harps* showed highest K_F (5.12 L/kg) due to highest organic content of (3.8%) (Lertpaitoonpan et al., 2009). Ibuprofen was adsorbed on volcanic soils of the agricultural field but adsorption capacities were low (Estevez et al., 2014).

4.3.4. Pharmaceutical removal by using combined methods

Many methods when used alone are not effective for pharmaceuticals removal, as detailed above from 4.1 to 4.3 sections. Better solution is to combine more than one method. For example, toxic chemicals are not good for microorganisms. If AOPs pre-treatment is applied it can decrease the rate of microorganism death and hence they will increase pharmaceutical degradation (Tekin et al., 2006). More work is needed in combining process research as they appear to be promising. Organic materials are sensitive to reverse osmosis so it could be good to use the treatment of carbon filtration. For AOPs method adsorption is mostly used treatment (Klavarioti et al., 2009). Amoxicillin is 99% removed by combining reverse osmosis and the Fenton process (Zhang, Ji and Xi, 2006). A combination study was done by using biological and solar photo-Fenton process for quinolone removal. Fenton alone removed 33% and when used combined removal was 95% and mineralization 90% (Sirtori et al., 2009). Nanofiltration and photocatalysis of quinolone showed that 66% antibiotic was removed and catalyst was removed by filtration (Molinari et al., 2006). For imidazole removal combination of adsorption and ozonation at the same time showed 90–100% degradation by ozonation and 10–20% by mineralization. Toxicity was decreased by 30% due to adsorbent present (Sánchez-Polo et al., 2008). Adsorption together with catalytic ozonation can be good for pharmaceuticals removal. Pharmaceutical is degraded by ozonation and adsorbent removes by products and remaining pharmaceuticals. Using a modified powder activated carbon, sulfamethoxazole was removed and adsorbent was used both for ozonation and adsorption (Akhtar et al., 2011). Tetracycline is effectively adsorbed and photocatalytically degraded by mesoporous BiOI microspheres (Hao et al., 2012).

5. Advantages and drawbacks of various removal processes

Removal efficiencies are different for pharmaceuticals by different water remediation methods. Plants for wastewater treatment are not made in the way to remove pharmaceuticals completely. Activated sludge treatment by sorption and biological degradation can remove pharmaceuticals. Sediments and sludge segregate pharmaceuticals having high sorption coefficient (Ahmed et al., 2017). A very good option for wastewater treatment is constructed wet land and provides good efficiencies. They are low cost sustainable answer to pharmaceuticals and large group contaminant removal (Matamoros et al., 2008). But major

disadvantage is large space need and retention times with constructed wet lands, urban area especially and they are pharmaceuticals hotspot areas (Matamoros and Bayona, 2007). Water contaminants ions are exchanged by resins ions in ion-exchange process. Ionization and softening are two highly famous ion exchange processes. Most microorganisms and organics are not much removed by deionization. Resins are attached by microorganisms hence provide culture for growth of bacteria and pyrogenic generation subsequently (Luo et al., 2014). In drinking water treatment chlorination is very important. However, pharmaceuticals with reactive functional groups are degraded by residual chlorine. Major problem is formation of degradation and chlorination by products. Chlorination by products in many cases are carcinogenic and highly toxic (José Rivera-Utrilla, Sánchez-Polo et al., 2013a, 2013b). Advanced filtration methods like micro, nano, and ultrafiltration remove pharmaceuticals effectively with high percentage of removal. Their applications are limited by high energy demand, high operational cost and membrane fouling (Ahmed et al., 2017). For many pharmaceuticals ozonation shows medium to high level removal efficiencies. If other reagents or catalysts like H_2O_2 are used in ozonation, removal efficiencies are increased (Esplugas et al., 2007). But its application has been hindered due to formation of by-products and high energy cost (Ahmed et al., 2017). Similarly, formation of oxidative by-products and high cost of other advanced oxidation process (O_3/UV , O_3/H_2O_2 , $Fe^{+3}/H_2O_2/UV$, Fe^{+2}/H_2O_2 , H_2O_2/UV) limit their use. Although they have high removal efficiencies (Ahmed et al., 2017). After reviewing many AOPs and oxidations, it is suggested that O_3/H_2O_2 is best for pharmaceuticals removal with minimum production of oxidation by-products (von Gunten, 2018). As advanced treatment, activated carbon is used with other water treatment operation and processes. In design of water treatment system placement of carbon to other component is a very good consideration.

Assessment of cost of different methods and advanced treatments are hardly found. Previously available reports for pharmaceuticals are site or effluent specific or work a particular pharmaceutical or treatment. Cost depends on many factors like construction processing, needs, recycling, availability, and issues of lifetime, labor cost, electricity price and other local problems. So, for a specific treatment facility cost will be different in variable locations with variable issues of treatment. Many factors are responsible for cost of individual adsorbent including processing needs, recycling, availability and lifetime issues. No literature is available on this information. With place and country, precursors cost also change. Cost depends on whether adsorbent is existing industry by-product; for example, existing bio-refinery part is pyrolysis and its value-added co-product is biochar, available at low cost. In water treatment for selection of adsorbent, cost evaluation is critical (Chakraborty et al., 2018; de Andrade et al., 2018). Adsorbent capacities must be included for different adsorbent cost comparison. An adsorbent with 3-fold higher price and 5-fold higher capacity may be best, with all other considerations equal (de Andrade et al., 2018). In order to do cost analysis of adsorbents, it's important to consider the cost of collection of precursors, decrease in size, carbonization process and activation finally. Availability of adsorbent, conditions of treatment, recovery of adsorbate easily, and reusability of adsorbent are effecting factors for adsorbent cost calculations (Chakraborty et al., 2018). When waste water plant receives pharmaceuticals, 20% decrease in effluent mineralization take place, which increase cost of treatment (Durán et al., 2018). It determines an adsorption treatment would be added or not. To determine adsorbent costs for pharmaceuticals removal from water, data is very limited. Adsorption typically needs less cost for application than many methods (Ahmed et al., 2015).

Sustainable and lower cost adsorbent development from waste materials would be economical. What else present in water to be removed will also determine the affordability of adsorption process. If wastewater treatment and multistep sewage are integrated with pharmaceuticals adsorption, then cost can be made interdependent with other steps. Using adsorption, cost of water treatment ranges from 10 to 200 US\$/million liters, depends on amount and kind of adsorbent used. Economically adsorption is better than electrolysis, electrodialysis, reverse osmosis and ion exchange. In these processes expected cost could be US \$450 per million liters (de Andrade et al., 2018). Treatment cost is increased due energy consumption (Monteil et al., 2019) and extensive chemical (Durán et al., 2018), use in advanced oxidation processes. To reduce cost it's suggested to combine chemical methods with biological process (Monteil et al., 2019). For decreasing operational costs promising technologies are radiation-based methodologies like solar-photo-Fenton processes. To further decrease cost ferrioxalate-aided methods are used which operate near pH 7 (Durán et al., 2018). For economic evaluation use of solar energy via photovoltaic panel provides an interesting case (Durán et al., 2018). Elimination of cyclophosphamide from hospital wastewater by ozonation, reports direct operation cost (oxygen supply + electricity) of 0.35 US\$/m³ (Ferre-Aracil et al., 2016; Lester et al., 2011). However use of electricity energy was equivalent to that required for trimethoprim degradation, where most cost economical was O₃ (Lester et al., 2011). Different pharmaceuticals will have variation in cost. Recent studies about antibiotics adsorption on different kinds of adsorbents described that most expensive were carbon nanotubes and biochars were least expensive (Ahmed et al., 2015). Different adsorbents approximate costs (US\$/kg) from literature are red mud (0.025), bagasse fly ash (0.02), biochar (2.65), bentonite (0.072), chitosan-based bioadsorbents (8–10), Metal organic frameworks (MIL-101) (5), commercial ACs (20–22), ion-exchange resin (150) and MOF (HKUST-1) (20) (de Andrade et al., 2018). These costs are suggestive only, as they depend on kind of precursor, local availability, conditions of treatment and disposal and re-generation. Excellent adsorbents are biochars due to relative low cost. Traditional granular activated carbons can be potentially replaced by biochars to reduce water treatment cost (Lin, Jiang and Xu, 2017). It was recently reported by Lin and others 20% higher sulamethoxazole was removed by biochar than activated biochar or granular AC. For pharmaceuticals removal biochars are projected promising. Biochar amount can be increased as side product of biofuels production (both syn gas and biooil) by using forest and energy crops or agriculture wastes, availability would increase, and price would drop. If biochar is applied as conditioner in agriculture and as soil amendment, it could enhance its use as adsorbent (Mohan et al., 2018).

6. Conclusion and future recommendations

Pharmaceutical contaminants are hazardous and polluting the ecosystem. They are responsible to cause acute and chronic diseases and affecting flora and fauna. Wastewater treatment plants are unable to mitigate these contaminants completely so these are moving into the drinking water supplies. As usage of pharmaceuticals has been increasing due to rise in population, so there is need to find the treatments for effective removal of pharmaceutical contaminants from environment. The employment of nanomaterials to mitigate pharmaceutical contaminants has been increasing in recent years. Nanostructured materials have large surface area, high porosity and high affinity for organic and inorganic molecules along with great catalytic characteristics. Adsorptive potential of nanomaterials made them highly valuable for effective removal of

pharmaceutical contaminants and this process does not led to formation of new toxic materials which are produced during other removal treatments. However huge number of adsorbents are required during adsorption which needs to regenerate and recovered contaminants also needs to be destroyed after recovery. Adsorptive removal of pharmaceutical contaminants is a lab scale practice largely also establish in small scale implementations. There is need to establish these practices on industrial scales. Application of different adsorbents for removal of pharmaceutical contaminants along with different optimized conditions, kinetics of adsorption have been reported in this review. All-important references covering all the research carried out on the analysis, remediation processes and suggestions for future concerns regarding pharmaceutical contaminants have been well documented.

Few important recommendations are listed below:

- 1) There is need to develop advanced methodologies for accurate determination of pharmaceutical contaminants in ecosystem
- 2) Strict rules and regulations should be established regarding release of industrial wastes into the environment
- 3) Use of green technologies should be instigated for manufacturing and development of pharmaceuticals.
- 4) Efficient and cost-effective technologies should be established for quick pharmaceutical mitigation on industrial scale

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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